

623 5Model: Short Flight-Test Guide

Updated June 23, 2026

1 Purpose and Build Approach

The current `623_5Model.ino` is a five-mode flight-test scaffold for MMAV v3. It keeps the present focus on glide testing while already reserving disabled code paths for GPS-triggered autorotation experiments.

The program is divided into five paths:

1. The IMU is sampled at 80 Hz, while the optional NEO-6M GPS supplies position and velocity at approximately 5 Hz.
2. A 15-state attitude and navigation estimator remains available for M3 glide stabilization and logging.
3. The M3 glide controller uses soft SMC with small wing authority and deadbands.
4. The M4/M5 autorotation logic is written into the code but disabled by compile-time switches.
5. Flight records are placed in a non-blocking queue and written to the SD card by a separate logger thread, so SD latency does not directly stop the control loop.

The first tests should still focus on M3 glide behavior. The M4/M5 code should be treated as planning and logging infrastructure until the autorotation mechanism is physically verified.

2 5Model Mode Definition

The flight sequence follows the latest five-mode structure:

Mode	Configuration	Main controls	Planner decision
M_0 carried/folded	Wings swept back, attached to quad	none	release location and altitude
M_1 folded acceleration	Wings still folded after release	maybe tail neutral	wait until safe deployment envelope
M_2 deployment transient	Wings sweeping forward	sweep actuator timing	survive deployment without tumble or stall
M_3 glide navigation	Wings open, tail available	differential wing incidence, optional tail	navigate toward autorotation entry region
M_4 autorotation transition	Wings flip/opposed	transition actuator	enter autorotation safely
M_5 autorotation landing	autorotation descent	servo3 descent-rate experiment	reduce landing error and impact speed

The current enabled path is only:

$$M_0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3.$$

M4 and M5 are present in the source code but disabled by default.

3 Current Enabled Behavior

The active code uses:

```
ENABLE_GPS_PHASE_TRANSITIONS = 0,    ENABLE_AUTOROTATION_PHASES = 0.
```

Therefore, GPS cannot currently trigger M4, and the aircraft cannot enter the autorotation-control code during normal tests.

The enabled mode behavior is:

- **M0:** unarmed or carried/folded; morph servo stays stowed.
- **M1:** after arming, wings remain folded for the folded-acceleration interval.
- **M2:** morph servo deploys the wing mechanism; SMC remains inhibited.
- **M3:** glide navigation begins; soft SMC is allowed.

4 Disabled M4/M5 Logic

The disabled future path is:

$$M_3 \xrightarrow{\text{GPS inside entry region for } N \text{ cycles}} M_4 \xrightarrow{\text{transition timer}} M_5.$$

The GPS trigger uses local NEU coordinates from the GPS origin:

$$d^2 = (p_N - p_{N,\text{entry}})^2 + (p_E - p_{E,\text{entry}})^2.$$

If

$$d \leq R_{\text{entry}}$$

for the required number of consecutive cycles, the disabled code can request M4.

In M4/M5, the current design deliberately avoids direction control. The first autorotation experiment is only descent-rate control:

- left blade servo returns to 90° ,
- right blade servo returns to 90° ,
- tail remains neutral,
- servo3 adjusts the autorotation angle,
- GPS down velocity estimates descent speed,
- GPS horizontal velocity approximates wind drift.

The M5 target is:

$$\delta_3 = \delta_{3,0} + k_v (v_{D,\text{filt}} - v_{D,\text{target}}),$$

where δ_3 is the servo3 target angle. The sign of this relationship must be measured because increasing servo3 angle may either increase descent rate or slow descent depending on the real mechanism.

5 Estimator and Filtering

The nominal estimator state is

$$\mathbf{x} = [\mathbf{q}, \mathbf{v}, \mathbf{p}, \mathbf{b}_g, \mathbf{b}_a],$$

and the 15-dimensional error state is

$$\delta \mathbf{x} = [\delta \boldsymbol{\theta}, \delta \mathbf{v}, \delta \mathbf{p}, \delta \mathbf{b}_g, \delta \mathbf{b}_a]^T.$$

For M3, the estimator provides roll, pitch, velocity, position, and bias estimates. For M4/M5, the intended first useful signals are simpler:

$$v_{D,\text{filt}}, \quad v_{\text{wind,approx}}, \quad \|\boldsymbol{\omega}\|_{\text{filt}}.$$

The approximate wind speed is not true air-relative wind. It is only GPS horizontal drift:

$$v_{\text{wind,approx}} = \sqrt{v_N^2 + v_E^2}.$$

6 M3 SMC Control

For each controlled M3 axis, the sliding surface is

$$s = \dot{e} + \lambda e.$$

The requested angular acceleration is

$$\dot{\boldsymbol{\omega}}_{\text{cmd}} = -\Lambda \boldsymbol{\omega} - K \text{sat}\left(\frac{\mathbf{s}}{\phi}\right) - \hat{\mathbf{d}}.$$

The wing command model is

$$\begin{bmatrix} \dot{p} \\ \dot{q} \end{bmatrix} = \underbrace{\begin{bmatrix} B_{00} & B_{01} \\ B_{10} & B_{11} \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} u_{\text{diff}} \\ u_{\text{sym}} \end{bmatrix} + \mathbf{d}.$$

For first M3 flights,

$$\mathbf{B} = \begin{bmatrix} B_{\text{roll}} & 0 \\ 0 & B_{\text{pitch}} \end{bmatrix},$$

with `ENABLE_ADAPTIVE_B=0`. The controller is intentionally soft: small attitude and rate errors are ignored so natural glide is not disturbed.

7 CSV Logging

The current file naming rule follows the 0620 test code. If:

`LOG_TAG = "0623",`

then files are created as:

`06_23_00.CSV, 06_23_01.CSV, 06_23_02.CSV, ...`

Each row records the current mode:

`mode_id, mode_name.`

The added autorotation log fields are:

`auto_vdown_filt, auto_wind_filt, auto_gyro_mag_filt, auto_morph_target_deg.`

These fields remain useful even while M4/M5 are disabled because they preserve a stable CSV structure for future tests.

8 Physical Quantities

Symbol	Unit	Meaning
$\mathbf{q}$	–	Body-to-world attitude quaternion
ϕ, θ, ψ	rad	Roll, pitch, and yaw angles
p, q, r	rad/s	Body roll, pitch, and yaw rates
$\mathbf{v}$	m/s	World-frame north, east, up velocity
$\mathbf{p}$	m	World-frame north, east, up position
$\mathbf{b}_g$	rad/s	Estimated gyroscope bias
$\mathbf{b}_a$	m/s ²	Estimated accelerometer bias
λ	s ⁻¹	Sliding-surface attitude convergence coefficient
K	rad/s ²	SMC switching acceleration gain
ϕ_{SMC}	rad/s	Boundary-layer width used to reduce chatter
u_{diff}	rad	Differential wing command for roll
u_{sym}	rad	Symmetric wing command for pitch
u_{tail}	rad	Optional tail pitch command, disabled by default
δ_3	deg	Servo3 morph or autorotation target angle
v_D	m/s	GPS down velocity, positive downward
$v_{\text{wind,approx}}$	m/s	GPS horizontal drift speed used as wind approximation
B_{ij}	s ⁻²	Angular-acceleration response per radian of wing command
Δt	s	Measured control-loop interval

9 Values That Are Not Yet Aircraft Measurements

Code parameter	Current value	Status and reason
B_ROLL_INITIAL	-6 s^{-2}	Assumed sign and magnitude; replace with measured roll effectiveness.
B_PITCH_INITIAL	$+6 \text{ s}^{-2}$	Assumed sign and magnitude; replace with measured pitch effectiveness.
B_TAIL_PITCH_INITIAL	$+6 \text{ s}^{-2}$	Placeholder only; measure before enabling tail pitch control.
PITCH_TARGET_RAD	0 rad	Placeholder; replace with measured safe glide trim attitude.
SERVO_CMD_LIMIT_DEG	5°	Low-authority M3 command range.
SERVO_MORPH_AUTOROTATION_*	80° to 130°	Placeholder range for disabled M5 descent-rate tests.
AUTOROTATION_DESCENT_SIGN	+1	Must be measured; the sign decides whether larger servo3 angle increases or decreases descent rate.
AUTOROTATION_ENTRY_*	local NEU meters	Placeholder GPS entry region for disabled M3-to-M4 transition.

10 Software Safety Configuration

The current intended configuration is:

- BENCH_MODE=0: timed automatic arming is disabled.
- ENABLE_CONTROL=1: M3 SMC is permitted only after hardware arming.
- ENABLE_ADAPTIVE_B=0: B remains fixed and diagonal.
- USE_TAIL_PITCH_CONTROL=0: D6 is fixed at tail trim.
- ENABLE_GPS_PHASE_TRANSITIONS=0: GPS cannot trigger M4.
- ENABLE_AUTOROTATION_PHASES=0: M4/M5 cannot execute.
- D2 uses INPUT_PULLUP; it must first remain open/high and then be connected to ground for at least four control cycles to arm.
- Opening the D2 circuit for at least four cycles disarms the controller and returns servos to safe commands.

11 Calibration and Flight-Test Procedure

1. Servo direction and range

Verify neutral PWM, differential wing motion, symmetric motion, morph-servo deployment, tail neutral, and mechanical clearance.

2. IMU axes

Verify that nose-up motion increases pitch and right-wing-down motion increases roll. Accelerometer and gyroscope mappings must use the same body-axis convention.

3. Timing and SD test

Run with SD logging and GPS connected. Require:

$$\text{missed_delta} = 0, \quad \text{control_hz} \approx 80.$$

4. Passive M3 glide test

Use estimator-only or very low-authority M3 control. Confirm that the aircraft can glide better than the earlier aggressive PD behavior.

5. Measure M3 control effectiveness

Measure B_{roll} and B_{pitch} from small, separate command pulses before trusting closed-loop M3.

6. Disabled M4/M5 preparation

Do not enable M4/M5 until M3 can reliably enter the desired region. Before enabling autorotation code, test servo3 angle versus descent rate using fixed-angle drops.

12 Flight Release Checklist

- Servo directions, neutral positions, linkage limits, and power supply verified.
- IMU axes and installation offsets verified.
- Hardware arm and disarm behavior verified using D2.
- CSV file naming confirmed as `MM_DD_ii.CSV`.
- Mode logging confirmed with `mode_id` and `mode_name`.
- M3 glide behavior reviewed before any M4/M5 test.
- M4/M5 compile switches confirmed disabled for normal glide tests.
- First test area, recovery method, and immediate disarm procedure prepared.

13 Update Log

June 15, 2026

- **Update 01:** Replaced the complementary-filter and PD concept with an 80 Hz 15-state ESEKF and boundary-layer SMC while retaining the non-blocking SD logging architecture derived from program 69.
- **Update 02:** Added optional four-pin u-blox NEO-6M support using UBX NAV-POSLLH, NAV-SOL, and NAV-VELNED messages, with absent or invalid GPS data represented by empty log fields instead of runtime errors.
- **Update 03:** Added measured-frequency, maximum-execution-time, missed-period, dropped-log, GPS, and ESEKF-reset telemetry so SD and estimator timing can be evaluated directly on the hardware.

June 17, 2026

- **Update 04:** Corrected angle wrapping, increased GPS local-coordinate conversion precision, corrected covariance initialization order, refreshed Euler angles after estimator reset, and removed a redundant Euler-angle extraction.
- **Update 05:** Added an initial arming delay, commanded neutral servos after an IMU read failure, and introduced a 16-cycle ESEKF settling interval before control could resume after a reset.
- **Update 06:** Reset the B -estimator derivative history after an IMU interruption so an old angular-rate sample could not create a false angular-acceleration impulse when IMU data returned.
- **Update 07:** Added gyroscope-calibration variance checking so vibration or changing angular motion during startup calibration causes calibration rejection instead of being stored as sensor bias.
- **Update 08:** Replaced the absolute B -matrix determinant threshold with a threshold relative to the diagonal product to reduce unsafe inversion of a nearly singular effectiveness matrix.
- **Update 09:** Required the B -matrix determinant to have the expected negative sign so excessive cross-axis estimates cannot pass the inverse-matrix check merely because their determinant magnitude is large.
- **Update 10:** Added acceleration-magnitude and mean-rate checks to startup calibration, tightened the acceptable gyro mean, and reduced the IMU read timeout from 20 ms to 8 ms to fit within the 12.5 ms control period.
- **Update 11:** Added separate bench/flight arming paths, a D2 hardware arm input, an IMU fault latch, an ESEKF fault latch, and a compile-time option that disables adaptive B estimation for initial flights.
- **Update 12:** Reordered the loop so IMU acquisition, ESEKF propagation, GPS correction, timing measurement, and SD logging continue while disarmed, preventing stale attitude from being used at the moment of arming.
- **Update 13:** Added `ENABLE_CONTROL` as an independent estimator-only mode so passive ESEKF/GPS/SD testing keeps both servos neutral without disabling sensor processing or logging.
- **Update 14:** Changed the default build to hardware arming, required the D2 input to be observed released before it can arm, and made removal of the arm signal actively disarm the controller and return both servos to neutral.
- **Update 15:** Added a shared control-state reset function that clears filtered commands, servo state, disturbance estimates, angular-acceleration history, and B -estimator readiness whenever control is armed, disarmed, or faulted.
- **Update 16:** Added four consecutive control-cycle confirmation for both D2 arm and disarm transitions so vibration or a one-frame electrical disturbance cannot momentarily reset and immediately restart the controller.
- **Update 17:** Unified IMU and ESEKF fault cleanup through the shared reset function, eliminating the earlier incomplete manual cleanup that left matched-disturbance state unchanged.

- **Update 18:** Reset the complete B matrix to the configured initial diagonal values on every control-state reset so a poor adaptive estimate cannot survive a disarm and contaminate the next arming cycle.
- **Update 19:** Cleared s_{roll} and s_{pitch} during control-state reset so disarmed and faulted log records no longer retain stale sliding-surface values.

June 18, 2026

- **Update 20:** Added a third D5 morphing servo that starts at the stowed angle, deploys to the configured morphing angle after one second, and logs its commanded angle and PWM without entering the SMC loop.
- **Update 21:** Matched the 0615 tested code style by adding a per-flight log tag, Core 0/Core 1 SD status messages, CSV stop markers, calibration output, per-side wing neutral/gain/incidence-trim servo mapping, and applied-incidence log columns.
- **Update 22:** Changed the D5 morphing servo deployment from 130° to 150° , giving a 60° motion from the 90° stowed position.
- **Update 23:** Added a fixed D6 tail-trim servo that is commanded to the configured neutral angle at startup, logged as tail angle and PWM, and left outside the ESEKF and SMC control loops.

June 19, 2026

- **Update 24:** Integrated the 0615 flight-test lesson by changing SMC into a neutral-biased soft stabilizer with small authority, deadbands, launch-delay intervention, explicit tail-trim fallback, optional tail pitch allocation, and added u_{tail} plus fault/armed status telemetry.
- **Update 25:** Reworked this guide to match the current four-servo MMAV v3 assumptions, clarify that GPS is not yet a landing-target controller, and group the update history by date.

June 23, 2026

- **Update 26:** Copied the current flight code into a new Arduino sketch named `623_5Model.ino` and renamed the boot identity to `623_5Model`.
- **Update 27:** Replaced the old CSV naming style with the 0620-style `MM_DD_ii.CSV` format and added `mode_id` plus `mode_name` to every CSV row.
- **Update 28:** Added the M0–M5 phase scaffold so the code records carried, folded acceleration, deployment transient, glide navigation, autorotation transition, and autorotation landing modes.
- **Update 29:** Added disabled GPS-triggered M3-to-M4 transition logic and disabled M4/M5 autorotation descent-rate control using GPS down velocity, horizontal drift approximation, gyro magnitude, and servo3 angle.